

XRComm FCS 8T8R Platform & Playback

Quick Start Guide

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1. Introduction

The XRComm FCS 8T8R Platform is a high-performance RF signal capture and distribution system designed for 8-channel transmit and 8-channel receive operations. It accepts raw IQ data from HSP ports and makes that data simultaneously available to logging, DSP, and IQ application consumers.

The platform also includes the XRComm 8T8R Playback engine, which streams raw I/Q signal data from a binary waveform file directly to a network interface at line rate (100 Gbps), supporting per-channel configurability.

2. Platform Components

Component	What It Does
<code>xrcomm_server</code>	The platform driver executable. Receives IQ packets, demultiplexes channels, and dispatches data to consumers simultaneously.
<code>xrcomm_grpc_ctrl</code>	The gRPC control server. Runs alongside the platform driver and exposes a network API for enabling/disabling modes and telemetry.
<code>playback_grpc_server.py</code>	The Python-based gRPC server for controlling waveform playback operations.
<code>xrcomm_nvme_loader</code>	Offline NVMe tool. Loads pre-recorded waveform files directly onto the NVMe drive so the platform can stream them without a network source.

3. Full Topology Block Diagram

[image]

4. Running the Platform Driver

Navigate to the platform driver directory and run the server as root:

```
cd XRComm_FCS_8T8R_platform/XRComm_platform_drivers/  
sudo ./bin/xrcomm_server
```

The platform server natively keeps the gRPC server active. Users should strictly interface with the platform via the gRPC Client running on the Client Computer.

5. Controlling the Platform over gRPC

The gRPC Client should be run on the **Client Computer**.

Endpoint Default: 127.0.0.1:50051. Use --host and --port flags to override.

RPC Reference (Platform)

RPC	Request	Purpose
SetDspMode	BoolRequest	Enable/disable DSP mode.
SetLogging	BoolRequest	Enable/disable NVMe capture.
SetIpMode	BoolRequest	Enable/disable per-channel IQ dispatch to applications.
GetStats	Empty	Combined pipeline counters and telemetry.
Shutdown	Empty	Gracefully stop the platform pipeline.

RPC Reference (Playback) The playback component provides its own gRPC service for controlling independent channels.

RPC	Request	Purpose
StartPlayback	StartPlaybackRequest	Provide a channel mask, start offset, duration in seconds, and sample_rate for specific channels.
StopPlayback	ChannelMask	Stop playback on specific channels.
GetActiveChannels	Empty	Returns an array of currently active channels.
GetPlaybackDiagnostics	Empty	Returns status metrics (equivalent to the JSON status file).

Appendix A: Environment Prerequisites

Supported Operating System: Ubuntu 22.04 LTS (64-bit) or Ubuntu 24.04 LTS.

Required Dependencies

Package	Version	Installation
DPDK	23.11.1	apt install dpdk libdpdk-dev
SPDK	24.01	Build from source
gRPC (Python runtime)	1.62.0	pip3 install grpcio grpcio-tools
CMake	3.16+	apt install cmake
GCC	11+	apt install build-essential

Hugepages The platform driver requires 2 MB hugepages pre-allocated before launch.

```
echo 4096 | sudo tee /sys/kernel/mm/hugepages/hugepages-2048kB/nr_hugep
sudo mkdir -p /dev/hugepages
sudo mount -t hugetlbfs nodev /dev/hugepages
```

NIC Binding Both receive NIC ports must be bound to the VFIO-PCI driver before starting the platform.

```
sudo modprobe vfio-pci
```

```
sudo dpdk-devbind.py --bind=vfio-pci 0000:xx:00.0 0000:xx:00.1
```